

6 (a) The table gives some information about structures found in chloroplasts.

Structure	Function	Other information
P	protein synthesis	some are 20 nm in size
Q	site of light-independent reactions	colourless
R	compartment for accumulation of hydrogen ions	largest are 4.35×10^{-4} mm long
S	storage granule	1 to 35 μ m in size

(i) Where in the chloroplast is RUBISCO active?

(1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** R
- ☐ **D** S

(ii) What is stored in S?

(1)

- ☐ **A** GALP
- ☐ **B** lipid
- ☐ **C** starch
- ☐ **D** sucrose

(iii) Where is GP formed?

(1)

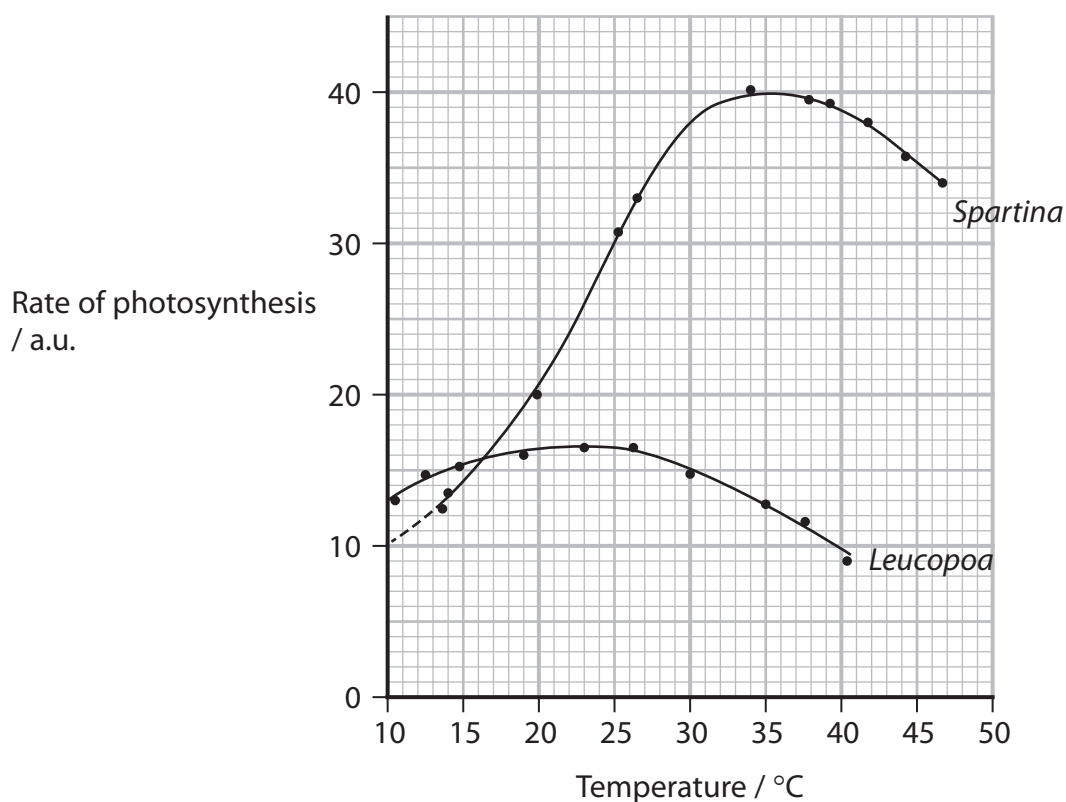
- ☐ **A** P
- ☐ **B** Q
- ☐ **C** R
- ☐ **D** S

(iv) Which is the correct order of size, smallest to largest?

(1)

- ☐ **A** P R S
- ☐ **B** P S R
- ☐ **C** R S P
- ☐ **D** S R P

(b) The graph shows the rate of photosynthesis in two types of plant, *Spartina* and *Leucopoa*, at different temperatures.



(i) Which could be the units for rate of photosynthesis?

(1)

- ☐ **A** mg CO₂ produced mm⁻² hr⁻¹
- ☐ **B** mg CO₂ produced mm⁻¹ hr⁻²
- ☐ **C** mg CO₂ used mm⁻² hr⁻¹
- ☐ **D** mg CO₂ used mm⁻¹ hr⁻²

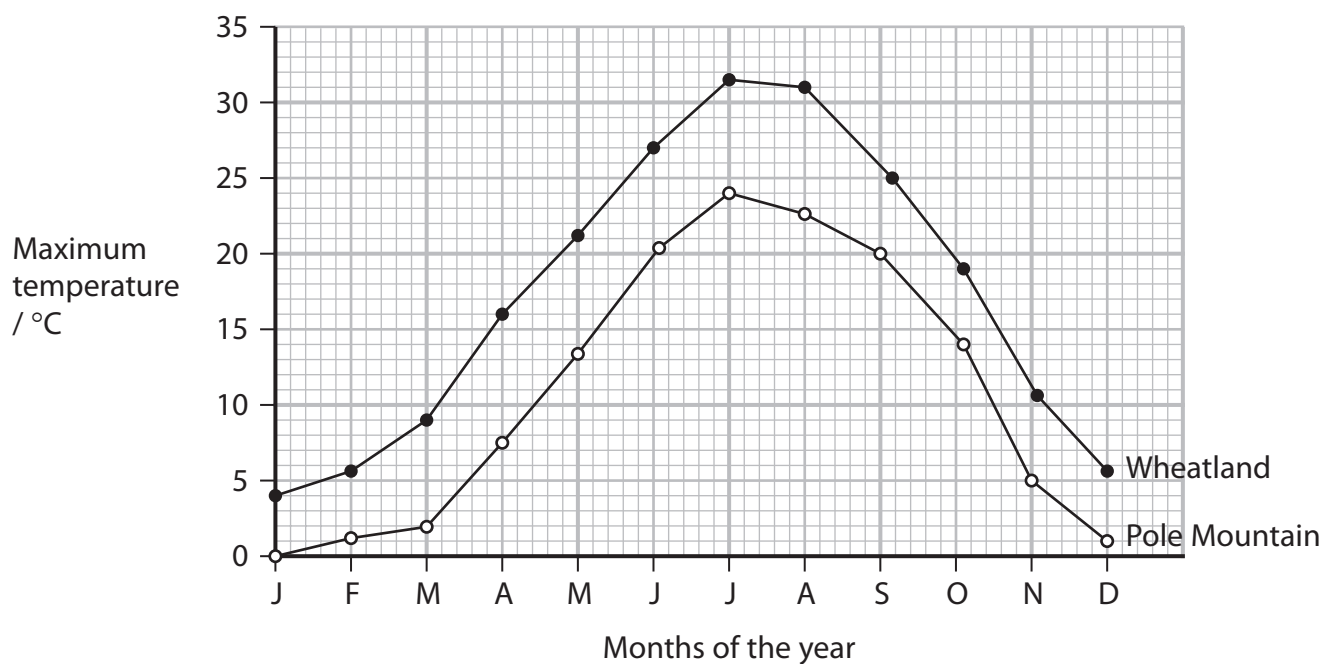
(ii) Compare and contrast the rate of photosynthesis of these two species of plant, at different temperatures.

(3)

(iii) Describe how a Q_{10} value for the rate of photosynthesis can be calculated, using this graph.

(2)

- (iv) The graph shows the maximum temperature, for each month, in two regions of North America: Wheatland and Pole Mountain.



Explain in which of these two regions of North America *Spartina* is most likely to grow.

Use the information in both graphs to support your answer.

(3)

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(Total for Question 6 = 13 marks)